

Amr Khalil

Unity Engineer | Simulation, XR, Digital Twins
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EXPERIENCE

Senior Unity Engineer

Oct 2024 – Present

BEDO Innovating Education

6th of October, Giza

- Developed Unity-based educational and industrial simulations, including baked fluid-behavior visuals.
- Built digital twin and digital shadow systems for PLC ladder logic and PID-controlled devices.
- Collaborated with instructional design, art, and mechatronics teams to deliver simulation products.
- Improved scalability using ScriptableObject architecture and dependency injection.

[Case Study](#)

Unity & AR Application Developer

Jun 2024 – Jun 2026

Ivris

Remote / Freelance

- Developed Ivris, a cross-platform Unity app for home interior visualization.
- Implemented runtime 3D model loading for furniture and room assets.
- Built Photon-based multiplayer room editing for real-time WebGL collaboration.
- Integrated Android/iOS AR previewing with PlayFab and a Render-hosted backend.

[Case Study](#)

Unity Virtual Reality Developer

Dec. 2023 – Oct 2024

Tanta University

Tanta, Gharbia Governorate

- Developed Meta Quest 2 VR applications for educational, lab, and medical training.
- Built a VR onboarding experience and viscosity lab simulation with interactive tools.
- Developed a multi-angle 360° surgical operation viewer with camera switching.

[Case Study](#)

SELECTED PROJECTS & AWARDS

Smart Irrigation System | IBM Call for Code 2019 Regional Winner, IoT, Machine Learning

- Built an agri-tech solution using IoT and machine learning to monitor plant health, reduce water usage, and support smarter irrigation decisions.

[Case Study](#)

Nescafé Ice Coffee VR Surfing Experience | Freelance, Unity VR

- Developed a VR surfing game for a marketing booth experience promoting Nescafé Ice Coffee.

[Case Study](#)

EDUCATION

German University In Cairo

Sept. 2014 – July 2019

Computer Science Engineering and Technology

Bachelor Grade: A – Overall Grade: B+

- Bachelor Project: Developed a Unity-based self-driving car simulation using reinforcement learning.
- Mentored bachelor students on Unity simulation, reinforcement learning, and self-driving car project topics.
- Delivered technical sessions on Unity fundamentals, physics, NavMesh, and simulation architecture.

TECHNICAL SKILLS

Languages: C# (Unity), JavaScript (WebGL .jslib), SQL, Java

Unity Development: Unity, WebGL, SOAP (ScriptableObject Architecture Pattern), PlayMaker

XR Development: VR, AR, Meta Quest 2, Auto Hand, 360 Video, Android, iOS

Simulation & Industrial: Digital Twins, Digital Shadows, PLC Ladder Logic, TIA Portal, PID Systems

Multiplayer & Backend: Photon, PlayFab, Node.js, Oracle Database, Render

Tools: Git, Visual Studio